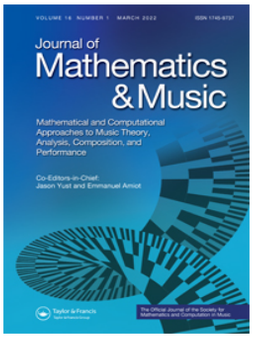

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Barberpole tempo illusions

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"Barberpole" tempo illusions are a family of auditory illusions based on the synchronization of faded rhythmic streams playing at different rates, often manufacturing experiences of seemingly eternal acceleration or deceleration. The forefather of all such illusions, based on layers whose rates are powers of two apart ("octaves"), was studied by Jean-Claude Risset in the late seventies and is now known as *Risset rhythm*. This article provides a mathematical framework for barberpole tempo illusions, generalizing Risset rhythms for arbitrary numbers of subdivisions, non-integer proportions, arbitrary rate modulation, and increasingly accelerating tempi. Furthermore, this article describes a new illusion of eternal rallentando/accelerando based on the full harmonic spectrum of rates. This construction shows that Risset rhythms are related to barberpole variable-rate polyrhythms. A notable application of the study of divisional structures that barberpole illusions underpin is the construction of bistable auditory figures (accelerating or decelerating depending on the stream being focused).

Keywords: Risset rhythms; auditory illusions; variable-rate polyrhythms; agogics; computer-aided composition

2010 Mathematics Subject Classifications: 00A65; 34A05
2012 Computing Classification Schemes: applied computing

1. Introduction

The study of auditory illusions lies at the interesting crossroad of psychoacoustics, composition, computer music and mathematics.

A class of illusions relies on the interference between a linear representation of a parameter (pitch, tempo) and a circular one (chroma, beat perception). We can manufacture situations in which the linear representation increases or decreases indefinitely, while the circular one regularly loops back: if we somehow manage to produce, at two different points in time, the same configuration of parameters, we obtain an ambiguous infinite motion, a bit like the rotating cylindrical signs, usually with coloured stripes, traditionally used in front of barbershops in the United States (Esqueda, Välimäki, and Parker 2015).

Auditory "barberpole" illusions are far from being *in vitro* experiments. They are timeshaping practices (Ghisi, Agostini, and Maestri 2020) which embody the quintessence of the archetype of infinity. The discrepancy between a linear and a circular perception challenges our intuition (Madison 2009), and we end up wondering how someone can produce a sequence of chords eternally going up on a piano keyboard, or a beat that indefinitely slows down. There is a magic side that acts powerfully not just on the interpretation of pitch and tempo, but also on the categories of space and time.

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This article provides a mathematical framework to account for several barberpole tempo illusions. The starting point are the well-known *Risset rhythms* (Risset 1978b), of which everything I mention can be seen as a generalization. I will use in these pages the valid formalism laid down by Stowell (2011). Section 2 briefly introduces Risset rhythms; Section 3 provides a general framework for arbitrary temporal subdivisions; Section 4 introduces a full-spectrum eternal accelerando or rallentando, by bridging down to the tempo domain a pitch circularity illusion designed by Deutsch, Dooley, and Henthorn (2008); Section 5 focuses on fabricating rate modulation functions to control the divisional time grids that barberpole tempo illusions underpin; Section 6 extends the Risset illusion to increasingly accelerating scenarios.

A secondary narrative all across the article is the quest for coincidences between the divisional onsets grid of an eternal accelerando and that of an eternal rallentando, in order to manufacture bistable auditory figures, appearing to both accelerate and decelerate, depending on the stream being focused.

An online repository[1] contains several audio examples along with a set of Max patches (Puckette 2002) which, by leveraging on the *bach* library (Agostini and Ghisi 2015), allow further personal exploration.[2]

2. Risset rhythms

The forefather of auditory barberpole illusions is the Shepard scale (Shepard 1964). Each note in the scale is synthesized through a set of partials spaced by octaves apart. Importantly, the amplitude envelope remains fixed as the notes vary: after one full octave, the frequency configuration reverts exactly to the initial state: partials fade in from silence in the lower range and fade out to silence in the higher one. Later, Jean-Claude Risset produced a continuous version of the same illusion (glissando).

The Shepard–Risset glissando conveys a very convincing perpetual upward or downward motion. The psychoacoustical roots of the illusion lie in the discrepancy between a circular tonal height and a linear spectral height (Risset 1978a): pitch is indeed a complex entity.

But the same consideration holds for tempo: the counterparts of the Shepard–Risset glissandi, as frequencies are pushed down in the realm of the infrasound, are called *Risset rhythms*. A Risset rhythm appears to be indefinitely accelerating or decelerating: under the hood, sped up or slowed down versions are constantly fading in and out. Layers are spaced by "octaves," each reading a source cell circularly at a rate of $1, 2, 4, 8 \ldots$ times the original (Figure 1). A fixed amplitude envelope, depending only on the rate, is assigned, so that a central rate is enhanced while its harmonics and subharmonics fade in and out. The only constraint on the source cell is that it be loopable: loops containing regular subdivisions will convey a clearer sense of shifting temporal grids (as the regularities will propagate through the layers, always in phase), but constructions with irregular events are also of interest for composers looking for less repetitive architectures. Differently from the case of the glissando, precise phase alignment is required to convey the illusion – which makes it much more delicate to construct.

As acknowledged by Jean-Claude Risset himself, the first rhythm behaving in this paradoxical way was synthesized by Kenneth Knowlton (Risset 1986); building upon it, Risset provided a more elegant construction along with several derived illusions, which he then exploited in his works, most notably in *Contre nature* (Risset 1997).

[1] https://github.com/danieleghisi/BarberpoleTempoIllusions.
[2] All patches are developed and tested on Max 8.1.5 (www.cycling74.com) and *bach* 0.8.1 (www.bachproject.net).

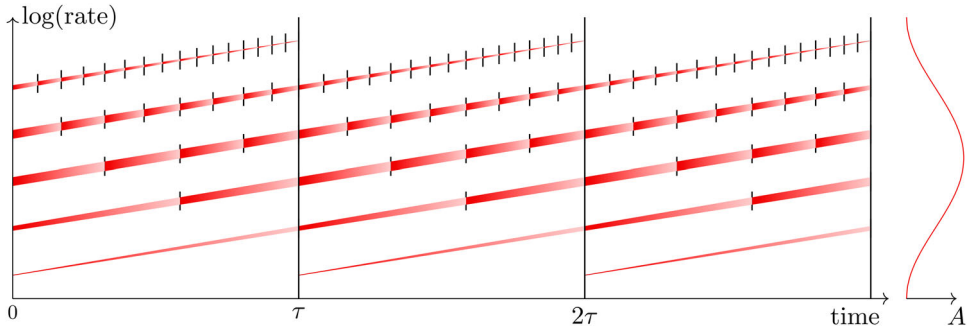


Figure 1. Visual representation of a Risset accelerando: a source cell (represented via the colour gradient) is read at different rates organized in streams or partials (the slanted lines), separated by "octaves" (each obtained, from bottom to top, by doubling the rate of the previous one). Partials double their frequencies in a time τ, joining seamlessly with the restart of the partial right above, since amplitudes depend only on the rate: bottommost partials fade in from silence and topmost partials fade out to silence. The structure is periodic, and the period τ is called "metabar." As the ticks in the diagram suggest, during a metabar, the fundamental partial reads one copy of the cell, the second harmonic reads two copies, the third harmonic reads four copies, and so on.

3. Arbitrary subdivisions

A first way to generalize Risset's construction is by removing the assumption of binary subdivisions and handling the case where each layer, after a period of τ (called the "metabar"), has increased its rate by a generic factor D. We start by focusing on the case in which either D or $1/D$ is an integer number.

Following the notational framework introduced by Stowell (2011), let T be the duration of a bar of source material (or "source cell"), whose time is denoted by t_L (for "logical time"). In contrast, t_E represents the actual physical time ("elapsed time"). The variable playback rate of the source material is denoted as

$$r = \frac{\mathrm{d}t_L}{\mathrm{d}t_E}.$$

We want primarily to control the rate as a function of the physical time, hence $r = r(t_E)$.

The fundamental idea is that we have N superposed layers $v = 0, \ldots, N-1$, each increasing its rate by a factor of D over the duration of the metabar ($0 \le t_E \le \tau$):

$$r_v(t_E) = r_v(0) D^{t_E/\tau}. \tag{1}$$

During the metabar, each layer repeats the original cell D^v times (if $D > 1$, during accelerandi) or $1/D^v$ times (if $0 < D < 1$, during rallentandi). We can gather the two cases by defining

$$\{D\} := \begin{cases} D & \text{if } D \ge 1, \\ 1/D & \text{if } 0 < D < 1. \end{cases} \tag{2}$$

We can integrate equation (1):

$$\int_{t_E=0}^{t_E=\tau} \frac{\mathrm{d}t_L}{\mathrm{d}t_E}\, \mathrm{d}t_E = \int_{t_E=0}^{t_E=\tau} r_v(0) D^{t_E/\tau}\, \mathrm{d}t_E.$$

Now notice that at $t_E = \tau$, $t_L = \{D\}^v T$; therefore, we get

$$\{D\}^v T = r_v(0)\tau \frac{D-1}{\ln(D)}$$

from which we obtain

$$r_v(0) = \frac{T}{\tau}\frac{\{D\}^v}{D-1}\ln(D). \tag{3}$$

Plugging equation (3) into equation (1):

$$\boxed{r_v(t_E) = \frac{T}{\tau}\frac{\{D\}^v \cdot D^{t_E/\tau}}{D-1}\ln(D).} \tag{4}$$

By integrating once more while imposing the boundary condition $t_L = 0$ when $t_E = 0$, we get the relationship between t_E and t_L:

$$t_L = T\frac{\{D\}^v \cdot (D^{t_E/\tau} - 1)}{D-1}. \tag{5}$$

or equivalently

$$t_E = \tau \log_D\left(1 + \frac{D-1}{T \cdot \{D\}^v} \cdot t_L\right). \tag{6}$$

As t_E goes from 0 to τ, t_L spans the interval $[0, \{D\}^v T]$ (i.e. $\{D\}^v$ repetitions of the source cell, read cyclically).

Sometimes (e.g. while using off-line timestretching algorithms) it is useful to relate the rate r_v to the logical time t_L instead. Plugging equation (6) into (4), we get the linear relation

$$r_v(t_L) = \frac{T}{\tau}\left(\frac{\{D\}^v}{D-1} + \frac{t_L}{T}\right)\ln(D). \tag{7}$$

Equations (4)–(7) hold both for accelerandi ($D > 1$) and rallentandi ($0 < D < 1$). For $D = 2$, one gets Stowell's formulae for a Risset accelerando, i.e.

$$r_v(t_E) = \frac{T}{\tau}2^{v+(t_E/\tau)}\ln(2), \quad t_L = T2^v(2^{t_E/\tau} - 1).$$

For $D = 1/2$, we get the formulae for the dual rallentando:

$$r_v(t_E) = \frac{T}{\tau}2^{v+1-(t_E/\tau)}\ln(2), \quad t_L = T2^{v+1}(1 - 2^{-t_E/\tau}).$$

A ternary example of accelerando ($D = 3$) is described by

$$r_v(t_E) = \frac{T}{2\tau}3^{v+(t_E/\tau)}\ln(3), \quad t_L = \frac{T}{2}3^v(3^{t_E/\tau} - 1)$$

and displayed in Figure 2.

In Risset rhythms, amplitudes depend uniquely on the rate r, with a mapping that can be more or less arbitrary, provided that layers fade in at lower rates and fade out at higher rates. Following Stowell (2011),[3] we use a cosine mapping:

$$A(r) = \frac{1}{2}\left(1 + \cos\left[\frac{2\pi}{b}\log_D\frac{r}{r_c}\right]_{-\pi}^{\pi}\right), \tag{8}$$

where r_c is the central rate (the one having maximum amplitude), b is a bandwidth in intervals of ratio D (e.g. in octaves for $D = 2$) and the square brackets represent the clipping of the argument in the interval $[-\pi, \pi]$.

[3] This also seems to correct a mistake in equation (3) of such article.

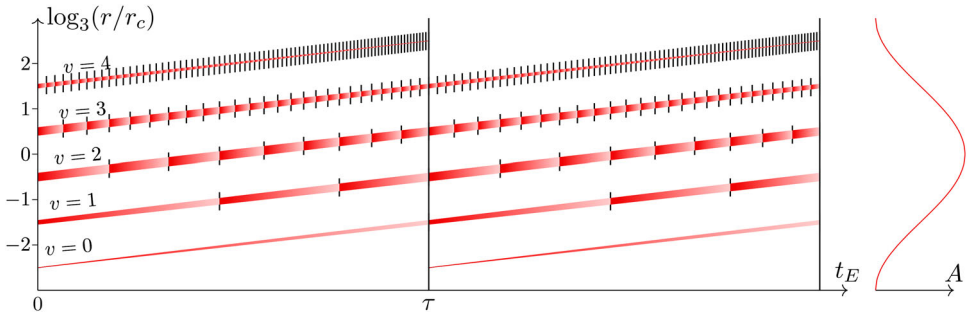


Figure 2. A representation of two instances of the metabar for a ternary Risset accelerando ($D = 3$). On the vertical axis, the tempo is represented logarithmically. Colour gradients offer a visual clue on the direction of the logical time; line size corresponds to amplitudes obeying a bell-shaped curve (displayed at right), centred around a principal rate r_c.

3.1. *Variable central rate r_c*

The central rate r_c controls the spectral height. By varying r_c over time, one can obtain interesting effects, such as a continuous rallentando that ends up being faster at the end than at the beginning. These illusions, already described by Risset (1997), exist verbatim in the extended framework. An alternative construction of a global accelerando within an infinite rallentando is via a continuous warping of the tempo, for instance by multiplying the playback rate by an exponential factor that does not loop at each metabar (while also shifting the central rate accordingly).

3.2. *A dual view on layers and non-integer {D}*

Instead of thinking of a stack of layers, each taking care of a portion of the frequency spectrum, we can switch to a dual view where each layer spans through the whole range of rates, starting from silence and fading to silence, the onset of each layer being calibrated to preserve the phase alignment. In this view, all layers play the same role.

Consider the case of an accelerando: a layer is a single sweep (the continuous slanted lines in Figure 2), indefinitely advancing the path of the root layer $v = 0$ as $t_E > \tau$ (always with the underlying assumption that the original cell is to be read cyclically):

$$r(t_E) = \frac{T}{\tau}\frac{D^{t_E/\tau}}{D-1}\ln(D). \tag{9}$$

The rate for the vth layer is simply the portion of curve between $t_E = v\tau$ and $t_E = (v+1)\tau$, and the number of repetitions of the source cell in the vth layer is

$$N_v = \frac{1}{T}\int_{v\tau}^{(v+1)\tau} r(t_E)\,\mathrm{d}t_E = \frac{R((v+1)\tau) - R(v\tau)}{T}, \tag{10}$$

with $R(t_E) = TD^{t_E/\tau}/(D-1)$ being a primitive of $r(t_E)$, i.e. up to a constant, the function yielding the logical time t_L. Our construction for an integer D works seamlessly because the number of cells in the vth layer is indeed

$$N_v = \frac{R((v+1)\tau) - R(v\tau)}{T} = \frac{D^{v+1}}{D-1} - \frac{D^v}{D-1} = \frac{D^v(D-1)}{D-1} = D^v.$$

Nonetheless, the interpretation of the barberpole structure as a sequence of full sweeps generalizes easily to the case where $\{D\}$ is not an integer: in this situation, τ is no longer a metabar (the layers no longer align eventually), but rather the offset between successive entries.

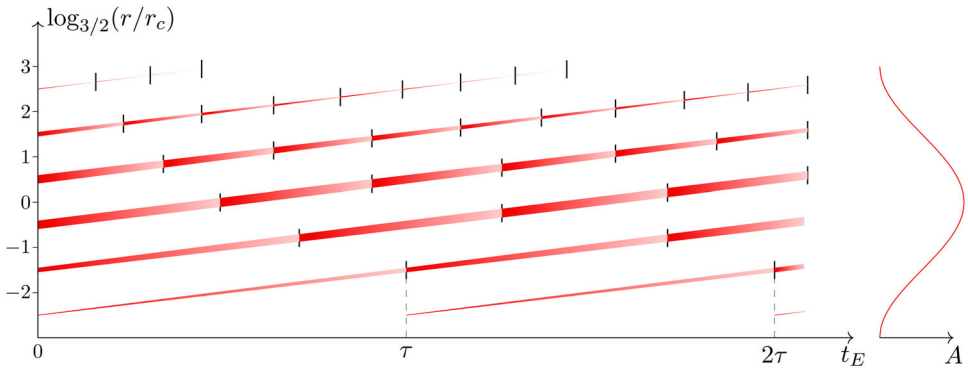


Figure 3. A representation of a generalized Risset accelerando (such as the one in Figure 2) with $D = 3/2$. The metabar no longer exists, and phase alignment follows 3:2 proportions.

The scenario with a rational $\{D\}$ corresponds to layers being arranged in rhythmical tunings based on intervals other than the octave. The simplest case is $D = 3/2$, corresponding to the stacking of hemiolas (see Figure 3):

$$r(t_E) = 2\,(\ln(3) - \ln(2))\,\frac{T}{\tau}\left(\frac{3}{2}\right)^{t_E/\tau}.$$

As $\{D\}$ moves away from simple proportions, or even becomes irrational, the phase alignment perceptually collapses and each sweep becomes more and more independent from the others.

3.3. *Divisional structure of Risset rhythms*

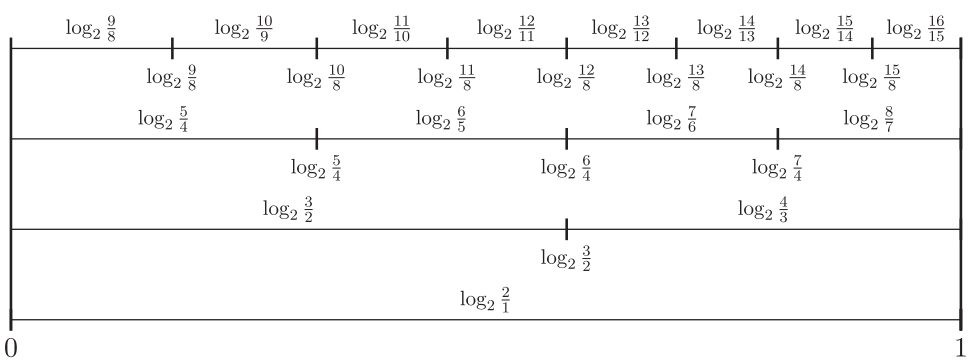
One can appreciate the elegant structures underpinning a Risset rhythm by visualizing the lengths of the repetitions of the cell as a division of a whole unit $T = \tau = 1$. Via equation (6), onsets follow the progression

$$\log_D\left(1 + \frac{D-1}{\{D\}^v}n\right), \quad \text{for } n = 0, \ldots, D^v,$$

and distances between them follow the progression

$$\log_D\left(1 + \frac{D-1}{\{D\}^v + n(D-1)}\right), \quad \text{for } n = 0, \ldots, D^v.$$

For instance, the structure of the first few layers of an ordinary Risset accelerando ($D = 2$) is the following:



Although the tree-like divisional representation may be reminiscent of fractals, there is no self-similarity in this structure.

If we think of the dual representation of layers as offset sweeps, the physical duration of the cells in a single sweep follows the simple progression

$$\log_2 \frac{2}{1}, \log_2 \frac{3}{2}, \log_2 \frac{4}{3}, \log_2 \frac{5}{4}, \ldots, \log_2 \frac{n+1}{n}, \ldots$$

For a generic $D > 1$, the progression is

$$\log_D \left(1 + \frac{D-1}{1+n(D-1)}\right), \quad \text{for } n = 0, 1, 2, \ldots$$

The representation for a rallentando $0 < D < 1$ is a flipped version of the corresponding accelerando $1/D$. For instance, the onsets $\log_2 \frac{m}{n}$ displayed in the previous $D = 2$ diagram are mirrored onto $1 - \log_2 \frac{m}{n} = \log_2 \frac{2n}{m}$.

4. A full-spectrum tempo illusion

Deutsch, Dooley, and Henthorn (2008) propose a model that conveys pitch circularity so that, differently from the Risset glissando, *all* harmonics, and not just octaves, show up. The authors gradually lower the pitch of a harmonic tone while reducing the amplitude of its odd harmonics, so that after a descent of one octave, the frequencies in the spectrum are the same as the initial ones. With a bit of manipulation on the intensities, one can then achieve a smooth transition.

There seems to be no reason why one shouldn't be able to carry this phenomenon "down" in the realm of infrasound: by taking care of phase alignment, we can construct an eternal rallentando that accounts for *all* rhythmic harmonics of a source cell (i.e. layers playing at double, triple, quadruple speed, and so on), superposing them and slowing them down, while fading out the odd harmonics and interpolating the amplitudes of the even ones.

Let us maintain our previous naming conventions, with the exception of the layers that are now assumed to be 1-based: $v = 1, \ldots, N$. The number v represents the harmonic number: by definition $r_v(t_E) = v \cdot r_1(t_E)$. For a smooth harmonic transposition, each layer must halve its rate across the metabar, so equation (1) still holds. However, the boundary condition is now that layer v must read v copies of the source cell during the metabar: $r_v(\tau) = vT$.

We get

$$\boxed{r_v(t_E) = v\frac{T}{\tau}2^{1-t_E/\tau}\ln(2)} \tag{11}$$

and integrating

$$t_L = 2vT(1 - 2^{-t_E/\tau}), \tag{12}$$

or equivalently

$$t_E = \tau \log_2 \frac{2vT}{2vT - t_L}. \tag{13}$$

As for intensities, we can assign them in arithmetic progression, layer v having some intensity level $-v\Delta L$, in decibels. During the course of one octave, odd layers fade out, while even layers v increase their intensity by $(v/2)\Delta L$, so that at the end of the vth layer the situation is exactly the same as the beginning of the $(v/2)$th.

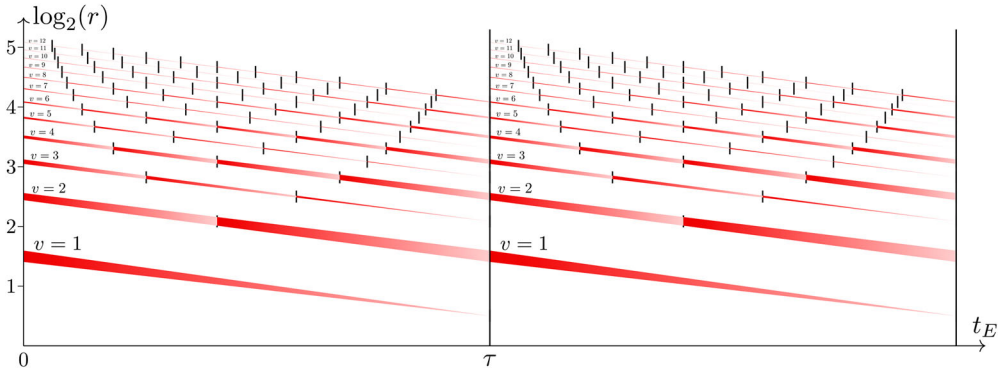


Figure 4. Visualization of a full-spectrum rallentando: logical time follows the colour gradient, intensity is mapped on the line width. As t_E approaches τ from the left, odd layers disappear and each even layer v joins seamlessly with the successive layer of index $v/2$.

Intensities can be controlled in other ways, as long as, at the metabar junction, odd layers fade to silence while every even layer v joins seamlessly with layer $v/2$. This yields the following "chains" of layers, each terminating with an odd harmonic:

$$\begin{aligned}
\ldots &\to 8 \to 4 \to 2 \to 1 \to \dagger \\
\ldots &\to 24 \to 12 \to 6 \to 3 \to \dagger \\
\ldots &\to 40 \to 20 \to 10 \to 5 \to \dagger \\
\ldots &\to 56 \to 28 \to 14 \to 7 \to \dagger \\
\ldots &\to 72 \to 36 \to 18 \to 9 \to \dagger \\
&\vdots
\end{aligned}$$

Ultimately, this rallentando extends Risset's construction (first chain) by adjoining non-dyadic copies, one for each odd number (all the other chains). Every harmonic is present because every positive number n can be factored as the product of an odd number and a power of 2 in a unique way: $n = (2m + 1)2^a$ for some integers $m \geq 0$ and $a \geq 0$. The nth tempo harmonic belongs to the mth chain (terminating with $2m + 1$) and is precisely its ath element from the right (both m and a are zero-based).

With these chains in mind, it is possible to introduce new musical material in layers at the left, provided that we propagate it to the right until the final odd layer is reached, where we can finally let it go. Chains are, to some degree, independent and can be processed separately, if needed.

The structure of the rallentando is displayed in Figure 4. Its dual accelerando counterpart can be obtained by flipping the diagram horizontally (or, equivalently, reversing the direction of the elapsed time).

Due to the presence of all harmonics in the mix, the rhythmic complexity of the rallentando is more pronounced than in Risset's case; this is however not without interest. One can consider this approach as belonging to the family of ideas of "rhythmical additive synthesis," dating back at least to Henry Cowell's visionary *New Musical Resources* (Cowell and Nicholls 1996), and then forged through the pioneering work of Nancarrow, Ligeti and Nunes among others. In the music of all these composers, inspired by nineteenth-century romantic hemiolas as well as African cross-rhythmic traditions, rational polyrhythmic structures emanating from non-octave relationships between layers are an invaluable richness rather than residual noise.

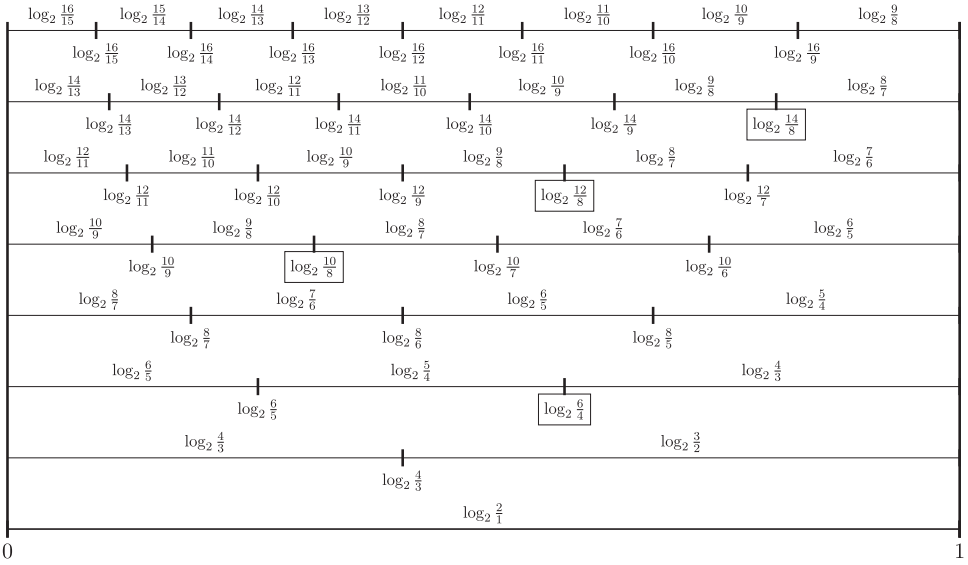
I have used this construction in my recent piece *On Rails*, a very long, continuously morphing Risset accelerando (first part) and rallentando (second part) of fragments of Ravel's *Boléro*, upon which variable-rate appearances of the main themes are embedded, often in the form of augmentation canons. Chains of layers corresponding to odd rhythmical harmonics (3rd, 5th and 7th) appear more and more pronouncedly, as a counterpoint to Ravel's orchestration which also begins to feature doublings of the theme with intervals of the harmonic series.

4.1. *Divisional structure*

By plugging $t_L = nT$ inside equation (13), we obtain the divisional structure of the full-spectrum rallentando. Layer v contains the following set of onsets

$$\tau \log_2 \frac{2v}{2v-n}, \quad \text{for } n = 0, 1, \ldots, v.$$

For $\tau = 1$, the superposition of the first eight harmonics is as follows:



Differently from the Risset case, intervals appear more than once. Importantly, odd layers contain all the onsets of the divisional structure of the ordinary Risset accelerando, the first few being highlighted with rectangles in the diagram above. In other words, the divisional structure of the Risset accelerando is a substructure of the full-spectrum rallentando.

4.2. *A bistable accelerando-rallentando*

This consideration suggests the possibility of constructing, by superposing a Risset accelerando on a full-spectrum rallentando, a bistable auditory image, for which it is ambiguous to state whether a given pattern is accelerating or slowing down indefinitely.

Of course, superposing a Risset accelerando to a Risset rallentando is a trivial task, but the absence of internal synchronicities makes for a very confusing rendition. A secondary narrative that will inform the following pages is concerned with how to clarify such rendition – which will prompt us to strike compromises between the smoothness of the tempo curves and the clarity of the divisional grids.

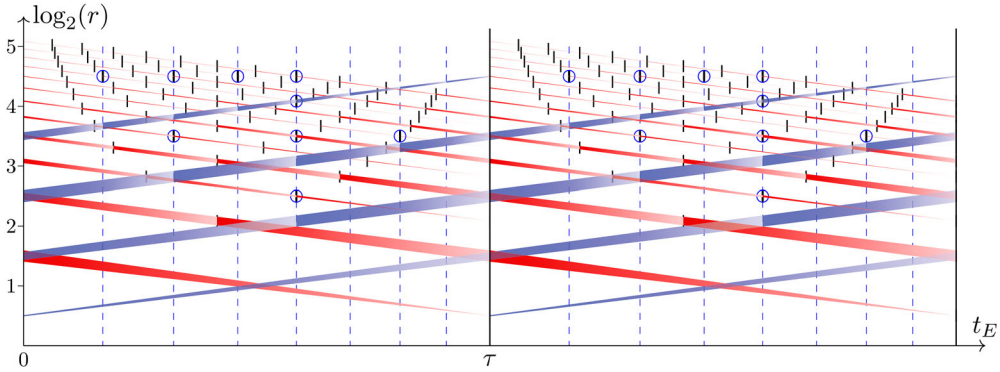


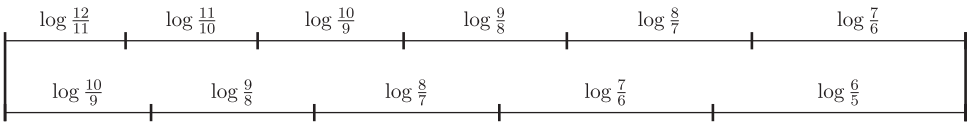
Figure 5. Superposition of the full-spectrum rallentando of Figure 4 (downward lines) with a standard Risset accelerando (upward lines). Dotted lines highlight the set of synchronicities across the two grids.

In any case, given that a possible path to clarify the mix is to find as many synchronicity points as possible, the full-spectrum construction may be seen as a step in the correct direction. That being said, producing a convincing bistable auditory image in this way is far from easy. For one thing, coincidence points are quickly high in the harmonic spectrum (Figure 5). To make matter worse, all odd layers of the full-spectrum rallentando fade to silence by design.

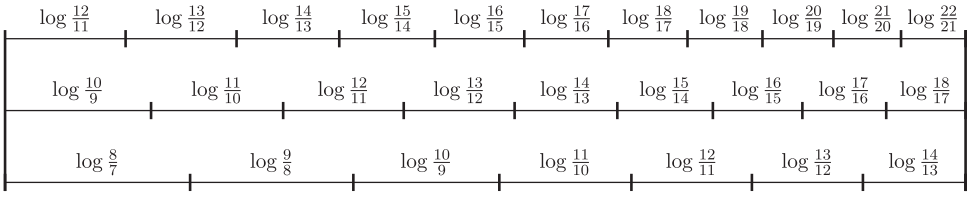
The best I was able to accomplish was by emphasizing the amplitudes of the odd layers of the full-spectrum rallentando around the synchronicities with the Risset accelerando. Some of the simplest connections (such as $t_E = \log_2 \frac{3}{2}$, $t_E = \log_2 \frac{5}{4}$, in addition to the obvious one at $t_E = 0$) are clear in the results. The rendering are somehow ambiguous, albeit admittedly not outstandingly so. An example is included in the online repository. In Section 5, I will try to provide a different approach to the problem.

4.3. *Variable-tempo polyrhythms*

Subsets of chains of the full-spectrum rallentando are, veritably, variable-tempo polyrhythms. For instance, a 5:6 polyrhythm in rallentando can be obtained via the following set of inter-onset intervals:



while a 7:9:11 polyrhythm in accelerando would be



All these layers must fade out, while their dyadic harmonics fade in, in order to ensure the eternal, barberpole construction as in the full-spectrum diagram of Figure 4. The base of the logarithm is irrelevant, as it only leads to a global stretch of the structure.

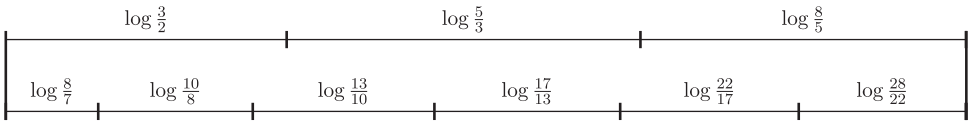
A $n_1 : n_2 : \ldots : n_M$ polyrhythm can be obtained by stacking M layers, where layer k has inter-onset intervals

$$\log \frac{k+1}{k}, \log \frac{k+2}{k+1}, \log \frac{k+3}{k+2}, \ldots, \log \frac{2k}{2k-1},$$

the sum of the inter-onset intervals in each layer yielding the constant value

$$\sum_{i=k}^{2k-1} \log \frac{i+1}{i} = \log \prod_{i=k}^{2k-1} \frac{i+1}{i} = \log \frac{2k}{k} = \log 2.$$

This construction works more generally for telescopic series, not necessarily in a constant accelerando or rallentando; for instance:



5. Tailoring the rate modulation and the divisional grid

Another way to generalize Risset rhythms is by modifying the rate function $\mathrm{d}t_L/\mathrm{d}t_E$. The exponential function is still a perfect choice for the job, but variations open a range of possibilities to control the divisional structure more finely.

Let us tackle the formalism from the beginning: layers revert to being 0-based and arranged in octaves or in intervals of ratio D, but now let us assume that a more general rate equation holds:

$$r_v(t_E) = r_v(0) f\left(\frac{t_E}{\tau}\right)$$

with $f\colon [0,1] \to \mathbf{R}$ being the rate modulation. Let us analyse the case of an accelerando: once we have an accelerando $f(t)$, we also have a dual rallentando $g(t) = 1/f(t)$. One may require, at the bare minimum One may require, at the bare minimum, that:

- f is continuous;
- $f(0) = 1$ and $f(1) = D$, i.e. the rates has a D-fold increase during the metabar. This ensures continuity in the chaining of layers;
- $\dfrac{\mathrm{d}f}{\mathrm{d}t} > 0$ for all $t \in (0,1)$, i.e. the rate grows at every point in time.

In the standard Risset case, $f(t) = 2^t$, $g(t) = 1/f(t) = 2^{-t}$.

Let $F(t)$ be a primitive of $f(t)$. The now familiar computation shows that

$$r_v(t_E) = \frac{T}{\tau} \frac{D^v}{F(1) - F(0)} f\left(\frac{t_E}{\tau}\right) \tag{14}$$

as well as

$$t_L = \frac{D^v T}{F(1) - F(0)} \left(f\left(\frac{t_E}{\tau}\right) - F(0) \right)$$

$$t_E = \tau F^{-1}\left(\frac{t_L}{T} \cdot \frac{F(1) - F(0)}{D^v} \right).$$

In particular, the onsets of the divisional structure for $\tau = 1$ are given by

$$F^{-1}\left(F(0) + \frac{n}{D^{\nu}}(F(1) - F(0))\right) \quad \text{for } n = 0, \ldots, D^{\nu}.$$

We can fix the degree of freedom on $F(x)$ by assuming that $F(0) = 0$, and the onsets become $F^{-1}(nF(1)/D^{\nu})$.

By tailoring $f(x)$, we control the rate more finely, but we have a looser control on the divisional grid. In contrast, by tailoring $F(x)$, we can fine-tune the divisional structure more directly, but we have less control on the behaviour of $f(x)$. In the following sections, we will briefly analyse some possibilities for manufacturing the two functions.

5.1. *Piecewise linear functions*

If we choose to control the rate modulation f directly, a piecewise linear function is a reasonable choice, as it is easy to treat: computing F is a trivial task and it is very easy to establish whether the desired properties of f hold. (On the other hand, clearly a piecewise linear function is a bad choice for F, due to f then being piecewise constant.)

5.2. *Polynomials*

Polynomial interpolation is unsuited for controlling either f or F, as it is hard to force monotony and convexity. Notice, in particular, that $F(x)$ must be differentiable and convex in $[0, 1]$. Since a non-negative weighted sum of convex functions is convex, we can think about representing $F(x)$ as a Taylor series with non-negative coefficients:

$$F(x) = a_1 x + a_2 x^2 + a_3 x^3 + \ldots \quad \text{with } a_i \geq 0 \; \forall \, i. \tag{15}$$

Any such $F(x)$ is absolutely monotonic in $[0, 1]$ (i.e. all of its derivatives are non-negative), but many simple convex functions are not included in the representation (for instance, $F(x) = x + x^2 - x^3 + x^4$). However, equation (15) still represents an important class of functions, including Risset's case via $a_i = (\ln(2))^{i-1}/i!$:

$$F(x) = \frac{2^x - 1}{\ln(2)} = x + \frac{\ln(2)}{2}x^2 + \frac{(\ln(2))^2}{6}x^3 + \frac{(\ln(2))^3}{24}x^4 + \ldots$$

Similar polynomials have the advantage of yielding straightforward calculations, but are hard to control via their inverse $F^{-1}(x)$, and hence, it is complicate to use them for the fine-tuning of divisional grids.

5.3. *Bézier curves*

As Bézier curves are contained within the convex hull of their control points, they constitute another natural choice for the manufacturing of either f or F. Importantly, Bézier splines provide local control, which is convenient when trying to adjust the behaviour of the divisional grid in the neighbourhood of a specific location.

Let us assume the dyadic Risset case of $D = 2$. We may model F with a quadratic Bézier interpolation of a set of n points $P_0, \ldots, P_n$. As $F(0) = 0$, one has $P_0 = (0, 0)$; also $P_n = (J, 0)$, with $F(1) = J$ for some $1 < J < 2$. Let $C_1, \ldots, C_n$ be the control points so that each triplet (P_{i-1}, C_i, P_i) determines the set of vertices of a quadratic Bézier curve. The simple case of $n = 1$ provides already some insight: C_1 can be pinned down by the constraints on the slope of the

tangents at P_0 ($F'(0) = 1$) and P_1 ($F'(1) = 2$); by varying a parameter $P_{n,y} = J$, we can explore rate modulations ranging from a constant function suddenly doubling at the end ($J = 1 + \varepsilon$), to a curve that doubles straight away ($J = 2 - \varepsilon$). Any intermediate point P_i can be added as a refinement of this curve. If we make the assumption that we add an internal point P_i so that the tangent at P_i has the same slope of the line between P_{i-1} and P_{i+1}, we can pin down all C_i's by imposing the following set of constraints, corresponding to the characterization of the slopes of the segments $\overline{P_{i-1}C_i}$ and $\overline{C_iP_i}$ for all $i = 1, \ldots, n$:

$$\begin{pmatrix} m_0 & -1 & 0 & 0 & 0 & 0 & \ldots & 0 & 0 \\ m_1 & -1 & 0 & 0 & 0 & 0 & \ldots & 0 & 0 \\ 0 & 0 & m_1 & -1 & 0 & 0 & \ldots & 0 & 0 \\ 0 & 0 & m_2 & -1 & 0 & 0 & \ldots & 0 & 0 \\ 0 & 0 & 0 & 0 & m_2 & -1 & \ldots & 0 & 0 \\ 0 & 0 & 0 & 0 & m_3 & -1 & \ldots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \ldots & m_{n-1} & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \ldots & m_n & -1 \end{pmatrix} \begin{pmatrix} C_{1,x} \\ C_{1,y} \\ C_{2,x} \\ C_{2,y} \\ C_{3,x} \\ C_{3,y} \\ \vdots \\ C_{n,x} \\ C_{n,y} \end{pmatrix} = \begin{pmatrix} m_0P_{0,x} - P_{0,y} \\ m_1P_{1,x} - P_{1,y} \\ m_1P_{1,x} - P_{1,y} \\ m_2P_{2,x} - P_{2,y} \\ m_2P_{2,x} - P_{2,y} \\ m_3P_{3,x} - P_{3,y} \\ \vdots \\ m_{n-1}P_{n-1,x} - P_{n-1,y} \\ m_nP_{n,x} - P_{n,y} \end{pmatrix}$$

with m_i's representing the tangent slope at point i:

$$m_0 = 1, \quad m_n = 2, \quad \text{and} \quad m_i = \frac{P_{i+1,y} - P_{i,y}}{P_{i+1,x} - P_{i,x}} \quad \forall\, i = 1, \ldots, n-1.$$

The advantage of designing F instead of f is that now we can control the qth divisional quantile, because it coincides with $F^{-1}(qJ)$, i.e. the x coordinate at which the function F assumes the value qJ. Of course, F must be invertible for that to happen; even more: f must be increasing, therefore F'' must be positive everywhere in $[0, 1]$. These conditions make for an extremely narrow wiggle room, riddled with painful compromises between customization of divisional grids and smoothness of acceleration curves. Figure 6 shows the functions and parameters used to produce the bistable audio examples included in the online repository.

If we push this framework farther still, conditions on f and F can be loosened to move towards a more general study of synchronicities across layers in variable-rate playback.

6. Increasingly accelerating tempi

By making the modulation function $f(x)$ dependent on the layer v, and namely by loosening the condition $f(1) = D$, we can attain an even more general class of rate modulation functions.

An interesting scenario is the following one: in principle, one could try to extend the idea behind Risset's barberpole structure to *increasingly* accelerating rhythms. We look for a single smooth function $r(t_E)$, with the idea of assigning to the vth layer the portion of function between $t_E = v\tau$ and $t_E = (v + 1)\tau$ (as we did in Section 3.2). Recall that the number of repetitions in layer v is given by equation (10).

Now consider a function $R(x)$ that grows faster than exponentially, such as $R(x) = T(D^{C^x - C + 1} - 1)$. For instance, for $D = C = 2$, one has $R(x) = T(2^{2^x - 1} - 1)$, yielding a number

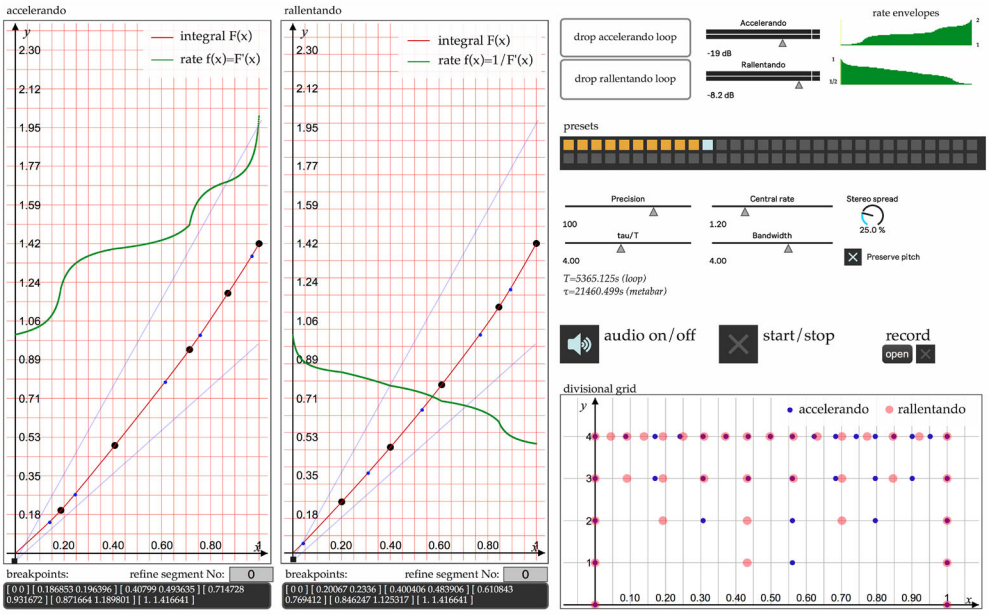


Figure 6. Screenshot of a Max patch. On the left: interface for tailoring two different functions $F(x)$, one for an accelerando (left) and one used for the dual rallentando (right): both are obtained via Bézier interpolation of a set of points (larger dots) with automatically computed control points (smaller dots). The winding curves that start at y = 1 show the rate modulation functions (i.e. the derivative in the case of the accelerando, and its reciprocal in the case of the rallentando). The horizontal grid lines subdivide the interval $[0, F(1)]$ into 16 equal steps. At the bottom-right corner, the divisional grid of the two figures is displayed.

of repetition for each layer of

$$
\begin{aligned}
v = 0: &\quad \frac{R(\tau) - R(0)}{T} = 1 - 0 = 1 \\
v = 1: &\quad \frac{R(2\tau) - R(\tau)}{T} = 7 - 1 = 6 \\
v = 2: &\quad \frac{R(3\tau) - R(2\tau)}{T} = 127 - 7 = 120 \\
&\quad \vdots
\end{aligned}
$$

and $2^{2^v-1}(2^{2^v} - 1)$ in general.

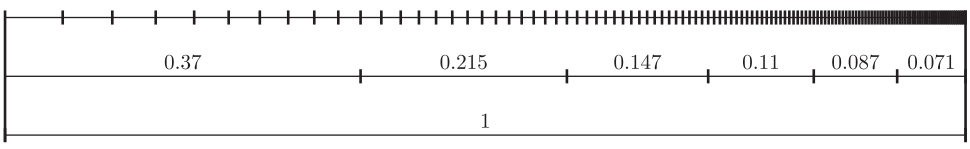
By differentiation, we get a rate function of

$$r(t_E) = \frac{T}{\tau}(\ln D)(\ln C)C^{t_E/\tau}D^{C^{t_E/\tau}-C+1}.$$

The usual calculations for the case $D = C = 2$ and $T = \tau = 1$ give

$$r(t_E) = (\ln 2)^2 \cdot 2^{t_E+2^{t_E}-1}, \quad t_L = 2^{2^{t_E}-1} - 1, \quad t_E = \log_2\left(\log_2(t_L + 1) + 1\right).$$

The first three layers of the divisional grid are



Due to the large distance between the rhythmical harmonics, the construction renders a result that is more akin to a sequence of synchronized and crossfaded accelerandi than a single eternal one. Nonetheless, the principle has a broader range of applications than the construction of auditory illusions and can be used, for instance, to design arborescent rhythmic structures with increasingly accelerating subdivisions.

Reducing D and C, at the expense of the transparency of the synchronizations, helps conveying more seamless transitions. In the online repository, examples are provided for $D = C = 2$ and for $D = C = 3/2$.

7. Conclusion and future work

In this article, I have provided a mathematical framework to account for barberpole tempo illusions with arbitrary subdivisions (including non-integer ones) and arbitrary rate modulation (including examples of increasingly accelerating rhythms), extending the work of Risset (1978b) within the notational framework of Stowell (2011).

With Cowell's idea of polyrhythms as an additive synthesis mechanism in mind, I have proposed an eternal rallentando or accelerando based on the full spectrum of "rhythmical harmonics" of a source cell. This illusion is the tempo counterpart of the one presented for pitch by Deutsch, Dooley, and Henthorn (2008). I have used similar barberpole constructions in my recent piece *On Rails*.

Finally, I have tried to study the divisional structures created by the multi-layer cell arrangements, taking advantage of synchronicities between the onsets of accelerandi and rallentandi to produce ambiguous auditory figures. The point of divisional schemes, however, goes beyond tailoring auditory illusions, and touches a broader range of applications including variable-tempo polyrhythms and the design of synchronicities across rate-dependent rhythmic streams. All these territories are promising areas of study for future research.

Importantly, this article has focused on the mathematics of barberpole tempo illusions, with compositional aspects in mind: future work may be dedicated to testing the new constructions in psychoacoustic experiments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

References

Agostini, Andrea, and Daniele Ghisi. 2015. "A Max Library for Musical Notation and Computer-aided Composition." *Computer Music Journal* 39 (2): 11–27.

Cowell, Henry, and David Nicholls. 1996. *New Musical Resources*. Cambridge, UK: Cambridge University Press.

Deutsch, Diana, Kevin Dooley, and Trevor Henthorn. 2008. "Pitch Circularity from Tones Comprising Full Harmonic Series." *The Journal of the Acoustical Society of America* 124 (1): 589–597.

Esqueda, Fabián, Vesa Välimäki, and Julian Parker. 2015. "Barberpole Phasing and Flanging Illusions." In *Proceedings of the International Conference on Digital Audio Effects*. Trondheim, Norway, 87–94.

Ghisi, Daniele, Andrea Agostini, and Eric Maestri. 2020. "Interactive Timeshaping of Musical Scores with Bach and Cage." In *Proceedings of the TENOR Conference*, 141–148.

Madison, Guy. 2009. "An Auditory Illusion of Infinite Tempo Change Based on Multiple Temporal Levels." *PLoS One* 4 (12): e8151.

Puckette, Miller. 2002. "Max at Seventeen." *Computer Music Journal* 26 (4): 31–43.

Risset, Jean-Claude. 1978a. *Hauteur et timbre des sons*. Technical Report. Rapport Ircam 11/78.

Risset, Jean-Claude. 1978b. *Paradoxes de hauteur*. Technical Report. Rapport Ircam 10/78.

Risset, Jean-Claude. 1986. "Pitch and Rhythm Paradoxes: Comments on 'Auditory Paradox Based on Fractal Waveform'." *The Journal of the Acoustical Society of America* 80 (3): 961–962.

Risset, Jean-Claude. 1997. "Rhythmic Paradoxes and Illusions: A Musical Illustration." In *Proceedings of the 1997 International Computer Music Conference (ICMC), 1997*, Thessaloniki, Greece, September 25–30, 1997, Michigan Publishing.

Shepard, Roger N. 1964. "Circularity in Judgments of Relative Pitch." *The Journal of the Acoustical Society of America* 36 (12): 2346–2353.

Stowell, Dan. 2011. "Scheduling and Composing with Risset Eternal Accelerando Rhythms." In *Proceedings of the 2011 International Computer Music Conference (ICMC), 2011*, Huddersfield, UK, July 31–August 5, 2011, Michigan Publishing, 474–477.